Building a Real-World Student–LLM Tutoring Dialogue Dataset for Feedback Analysis and Model Improvement

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Abstract

This project builds a dataset of authentic student–LLM tutoring dialogues collected from real educational interactions in Computer Science and Mathematics settings. The dataset contains approximately 100 multi-turn and single-turn conversations from around 20 students and is annotated with learning scenarios, learning goals, prompt quality, and tutoring response quality. We develop a pedagogically motivated scoring framework that evaluates not only correctness, but also coherence and educational guidance. In addition, the project investigates tutoring strategies, follow-up questioning behavior, and multi-turn confusion repair in student–LLM interactions. Through this dataset and analysis framework, we aim to better understand how students naturally use LLMs as tutoring tools and how pedagogical quality can be evaluated in real-world educational settings.

1 Introduction

Large Language Models (LLMs) are increasingly used as tutoring tools, yet their effectiveness as pedagogical agents remains underexplored. Although prior work studies LLMs in educational settings, much of it relies on simulated or otherwise structured interactions that may not reflect how students naturally use these systems. Existing datasets also rarely focus on authentic college-level Computer Science and Mathematics help-seeking conversations, despite these subjects being among the most common and important domains in which students rely on AI assistance. In these fields, tutoring interactions are often highly dynamic, involving debugging, multi-step reasoning, clarification requests, iterative follow-up questions, and conceptual repair across multiple turns.

Our project therefore aims to build a dataset of authentic student–LLM tutoring dialogues drawn from real problem-solving interactions. Unlike benchmark-style datasets or simulated tutoring systems, our dataset captures interactions initiated by real students asking genuine questions related to coursework, assignments, exams, and research projects. The dataset focuses primarily on Computer Science and Mathematics domains and emphasizes pedagogical quality in addition to correctness.

The main research questions are:

- How do students naturally use LLMs in tutoring interactions?
- What learning goals and learning scenarios appear most frequently in student–LLM tutoring dialogues?
- How does prompt quality relate to response quality?
- What types of feedback and pedagogical strategies do LLMs provide?
- How do LLMs respond to follow-up questions and repair student confusion across multiple turns?

To study these interactions, we develop an annotation and evaluation framework for analyzing tutoring behavior, learning goals, and pedagogical quality in student–LLM dialogues. Our analysis focuses not only on correctness, but also on how effectively LLMs support learning through explanation, guidance, clarification, and multi-turn interaction.

2 Methods

2.1 Datasets

We collected approximately 100 student–LLM tutoring dialogues from around 20 students, primarily from Barnard College and Columbia University. The dataset mainly focuses on Computer Science and Mathematics learning scenarios, including programming assignments, debugging tasks, conceptual learning, lecture review, and exam preparation.

Since many students frequently use screenshots or image-based prompts when interacting with LLMs, we converted screenshot-based conversations into text during preprocessing and data cleaning. For example, screenshots containing code, equations, or problem statements were manually transcribed into textual dialogue format. This preprocessing step made the conversations easier to store in structured JSON format and allowed both human annotators and LLM-based annotators to process the dialogues more consistently during downstream analysis.

The collected dialogues were first organized into Word documents before being converted into structured JSON files. During this process, we initially attempted to use ChatGPT to automatically convert the dialogues into JSON format. However, we observed that the model sometimes summarized or abbreviated portions of the conversations even when explicitly instructed to preserve the full dialogue content. To reduce information loss, we refined our prompting strategy, emphasized full-text preservation more explicitly, and in some cases manually copied the dialogue text directly into the chat interface rather than uploading Word documents. We also experimented with Claude for dialogue conversion, which preserved long-form conversations more reliably. These preprocessing and cleaning steps were important for maintaining dialogue completeness and annotation consistency throughout the dataset construction pipeline. We additionally performed data cleaning to improve formatting consistency and annotation quality. In particular, we deliberately excluded dialogues containing large amounts of graphs or highly visual outputs because these are difficult to reliably represent in JSON format and harder for automated annotation systems to process consistently.

The final dataset contains both human-annotated and AI-assisted annotated dialogues. Specifically, 60 dialogues were manually annotated by our team, while 40 additional dialogues were annotated with AI assistance. For the AI-assisted annotation process, we first provided manually annotated examples to the LLM in order to demonstrate the annotation framework and labeling logic before generating annotations for additional dialogues.

Statistic	Value
Total dialogues	$\sim$100
Students	$\sim$20
Human-annotated	60
AI-assisted annotated	40
Primary domains	CS / Math
Multi-turn dialogues	70%
Single-turn dialogues	30%
Storage format	JSON

Table 1: Basic dataset statistics.

Each dialogue is stored as a structured JSON object containing the conversation content, annotation labels, prompt quality scores, response quality scores, and evaluator notes. An abbreviated example is shown below:

The corresponding annotation labels classify this interaction as an Assignment-related dialogue with the learning goal of Concept Explanation. The dialogue additionally includes prompt quality and response quality scores based on our pedagogical evaluation framework.

2.2 Models and Approach

Our project focuses on analyzing authentic tutoring interactions rather than training a new LLM from scratch. The dialogues in our dataset were collected from student interactions with several widely used general-purpose LLM systems, including ChatGPT [OpenAI, 2023], Gemini [Team, 2023], and Claude [**?**].

For ChatGPT, the collected dialogues include interactions with GPT-4.5, GPT-5, and GPT-5.5 models, including both Thinking and Instant modes. For Gemini, the dataset primarily contains interactions with Gemini 3 in both Fast and Thinking modes. The dataset also includes dialogues interacted with Claude Sonnet 4.6.

All of these systems are transformer-based large language models designed for general-purpose conversational and reasoning tasks. These models were selected because they are among the most commonly used AI assistants in real student learning environments. Rather than evaluating highly specialized tutoring systems, our project focuses on the general-purpose LLMs that students already use informally

Listing 1: Abbreviated example of an annotated tutoring dialogue.

```
{
  "conversation_id": "D46",
  "llm": "GPT",
  "conversation": [
    {
      "role": "student",
      "content":
      "Why do you need to
      reaccess memory after
      overwriting a register?"
    },

    {
      "role": "llm",
      "content":
      "The response explains
      that memory still stores
      the value while the
      register does not."
    }
  ],

  "annotations": {
    "Scenario":
    "Assignment",

    "Goal":
    "Concept Explanation",

    "prompt_quality": {
      "Clarity": 4,
      "Specificity": 4,
      "Context": 3,
      "Total": 11
    },

    "response_quality": {
      "Coherence": 5,
      "Correctness": 5,
      "Guidance": 5,
      "Total": 30
    }
  }
}
```

for college-level learning.

In addition to collecting the dialogues, we developed a structured annotation framework for analyzing tutoring behavior and pedagogical quality. To establish annotation consistency, 60 dialogues were manually annotated first in order to develop annotation guidelines and identify common tutoring patterns. The remaining dialogues were then annotated with ChatGPT 5.5 based on the established annotation framework.

2.3 Experiments

To systematically analyze tutoring behavior and pedagogical quality, we developed a two-level annotation framework for the collected dialogues.

First, each student's prompt is categorized according to the learning scenario:

- Assignment Help
- Exam Preparation
- Lecture Learning
- Project or Research Support
- Not Clear

Second, each student's prompt is labeled according to the learning goal:

- Concept Explanation
- Step-by-Step Guidance
- Answer Clarification
- Direct Answer Generation
- Strategy or Hint Seeking
- Debugging / Error Fixing

To establish annotation consistency, 60 dialogues were manually annotated first in order to develop annotation guidelines and identify common tutoring patterns. The remaining 40 dialogues were then annotated with ChatGPT 5.5 based on the established annotation structure and labeling framework.

We additionally evaluate both student prompts and LLM responses using numerical scoring systems.

Prompt quality is evaluated according to three criteria:

- Clarity of Intent
- Specificity and Precision
- Context Provision

Each criterion is scored on a scale from 1 to 5, and the total prompt quality score is computed as the sum of the three categories.

Response quality is evaluated according to:

- Coherence
- Correctness
- Guidance Quality

Unlike prompt quality, response quality uses a weighted scoring strategy:

$$S_r = 2C + R + 3G$$

where C denotes coherence, R denotes correctness, and G denotes guidance quality.

We intentionally assign the highest weight to Guidance Quality because the primary goal of tutoring is not simply to provide correct answers, but to support student understanding and learning progression. A response may be factually correct while still being pedagogically ineffective if it lacks explanation, scaffolding, or meaningful guidance.

Coherence is assigned the second-highest weight because responses must remain logically organized and understandable in order to function effectively as tutoring interactions. Correctness remains important, but receives a smaller weight because correctness alone does not necessarily indicate strong tutoring behavior. Therefore, our research focuses more heavily on the pedagogical quality of tutoring interactions, including guidance, explanation structure, and learning support.

Using the annotated dataset, we conduct several forms of pedagogical and interaction-based analysis. These include examining the tutoring strategies commonly used by LLMs, studying how models respond to follow-up questions, and analyzing how they repair student confusion across multiple turns. We also analyze relationships between prompt quality and response quality, compare tutoring behaviors across different learning scenarios and learning goals, and investigate how interaction patterns influence pedagogical effectiveness.

3 Results and Analyses

3.1 Dataset Overview

The final dataset contains 162 labeled student turns drawn from 97 conversations. Of these, 64 turns are human-annotated by our team and 98 turns were labeled using GPT-4o-mini few-shot prompting applied to the 57 previously unannotated conversations. Table 2 shows the combined goal distribution. Concept Explanation (59%) dominates, reflecting that students most commonly seek explanations of concepts they have not fully understood. Answer Clarification (17%) and Step-by-Step Guidance (12%) are the next most frequent, while Debugging/Error Fixing (7%) and Direct Answer Generation (4%) are less common.

For learning scenarios, Lecture Learning (46%) is the most common context, followed by Not Clear (35%), Assignment Help (14%), and Project/Research Support (5%). The affective label distribution across the 64 human-annotated turns shows that 55% of follow-up messages are neutral, 25% satisfied, and 20% unsatisfied — indicating roughly one in five student follow-up messages signals remaining confusion or dissatisfaction after the LLM's response.

Learning Goal	Count	%
Concept Explanation	96	59%
Answer Clarification	28	17%
Step-by-Step Guidance	20	12%
Debugging / Error Fixing	11	7%
Direct Answer Generation	7	4%

Table 2: Learning goal distribution across 162 labeled turns (64 human-annotated, 98 GPT-labeled).

3.2 Prompt and Response Quality

Across the 64 human-annotated turns, the mean normalized prompt quality score is 0.757 ($\sigma = 0.216$), and the mean normalized response quality score is 0.810 ($\sigma = 0.192$). Both are relatively high, consistent with the finding that modern LLMs produce coherent and correct responses even to moderately specified prompts. The slightly higher mean response quality suggests that LLMs compensate for underspecified student prompts.

The Pearson correlation between prompt quality and response quality is $r = 0.228$, indicating a weak positive relationship. While higher-quality prompts are associated with marginally better responses, the relationship is not strong, suggesting that response quality is only loosely coupled to prompt specificity. This may reflect the robustness of modern LLMs across a wide range of prompt quality levels.

3.3 Goal and Satisfaction Classification

To evaluate whether annotations can support automated intent detection, we trained three classifiers on the goal prediction and satisfaction prediction tasks: (1) TF-IDF with logistic regression, combining bag-of-ngrams features with normalized quality scores; (2) a Sentence-Transformer MLP using `all-MiniLM-L6-v2` embeddings concatenated with quality scores; and (3) a TextCNN with learned token embeddings.

For goal prediction, the full 162-row dataset was split 8:1:1 stratified by class (128 train, 17 val, 17 test). The training set includes both human-annotated and GPT-labeled turns; the test set similarly contains a mix of both sources. For satisfaction prediction, LOOCV was applied over the 64 human-annotated turns. Results are reported in Table 3.

Task	Model	Acc.	Macro-F1
Goal (5-class)	TF-IDF + LogReg	**0.706**	**0.531**
	Sentence-MLP	**0.706**	0.524
	TextCNN	0.588	0.217
Satisfaction (3-class)	TF-IDF + LogReg	**0.578**	**0.525**
	Sentence-MLP	0.531	0.446
	TextCNN	0.547	0.298

Table 3: Classification results. Goal: majority baseline $= 0.588$ (17-sample test set, $n = 162$). Satisfaction: majority baseline ≈ 0.547 (LOOCV, $n = 64$).

For goal prediction, TF-IDF and Sentence-MLP both achieve 70.6% accuracy, outperforming the majority baseline of 58.8% by 11.8 percentage points. Adding GPT-scored response quality features (coherence, correctness, guidance) meaningfully improved macro-F1 for both models — TF-IDF reaching 0.531 and Sentence-MLP 0.524 — indicating that quality scores help distinguish minority classes beyond what text alone can capture. TF-IDF achieves the best macro-F1 (0.531) and correctly classifies Debugging/Error Fixing with perfect precision and recall. TextCNN, which uses text only and cannot leverage quality scores, drops to 58.8% accuracy and macro-F1 of 0.217, collapsing largely to the majority class.

For satisfaction prediction, TF-IDF with logistic regression achieves the highest accuracy (57.8%) and macro-F1 (0.525), both above the 54.7% majority baseline. The simpler TF-IDF model outperforming neural alternatives is expected when training data is scarce: n-gram features generalise more directly from short text than learned representations, which require more examples. TextCNN's low macro-F1 (0.298) despite reasonable accuracy reveals class collapse — it predominantly predicts the majority class (neutral).

3.4 Confusion Matrices

Figures 1 and 2 show the confusion matrices for the goal and satisfaction tasks respectively. For goal prediction, all three models classify Concept Explanation reliably (the dominant class), while Debugging/Error Fixing and Direct Answer Generation are consistently missed due to their small support. The Sentence-MLP shows the most balanced recall across classes. For satisfaction prediction, TF-IDF captures all three classes to some degree, while TextCNN collapses almost entirely to the neutral majority class.

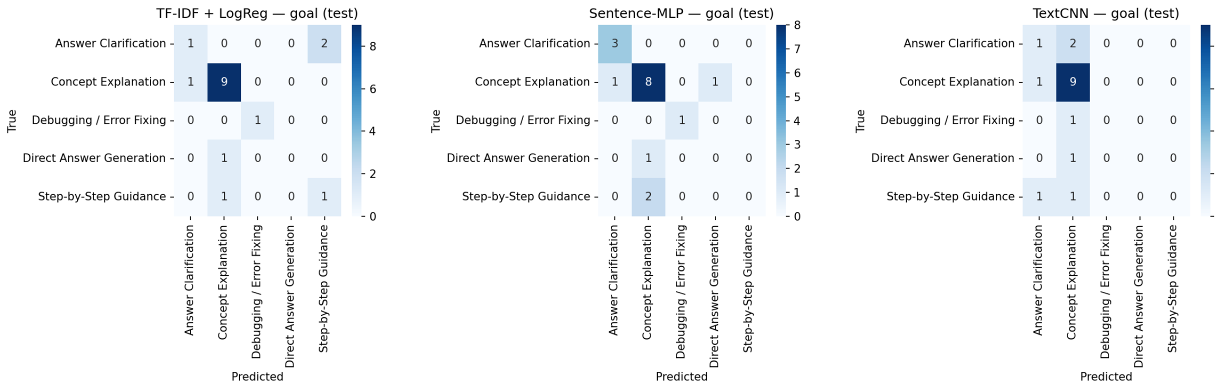
3.5 Limitations

The goal classification test set contains a mix of human-annotated and GPT-labeled turns, since the 8:1:1 split was applied to the full 162-row dataset. While GPT-4o-mini is a reliable labeler for intent classification tasks, this means evaluation is partially against silver-standard rather than gold-standard labels. For satisfaction prediction, LOOCV is applied solely over the 64 human-annotated turns, making that evaluation fully grounded in human judgement. Additionally, Debugging/Error Fixing and Direct Answer Generation remain underrepresented even after GPT-assisted labeling, limiting per-class performance on those categories.

4 Related Work

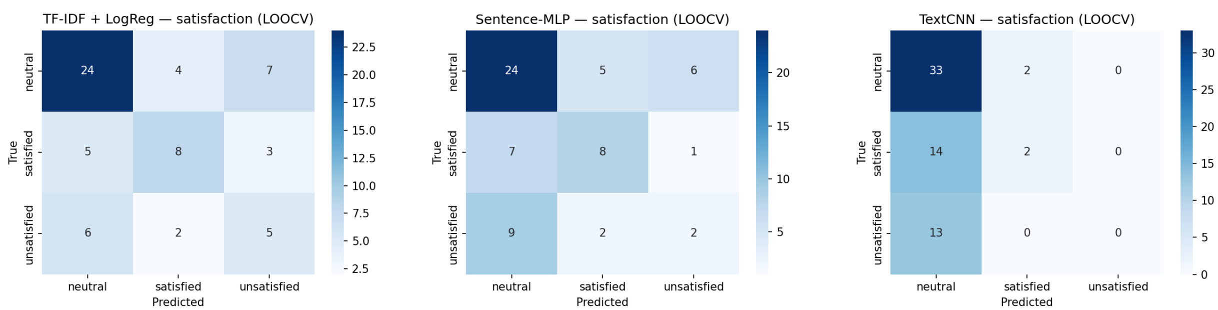
Prior work on LLM tutoring spans several strands, including tutoring dialogue datasets, analyses of real student–AI interactions, and methods for improving tutor behavior from dialogue data. First, tutoring research has long emphasized that effective learning support depends on multi-turn interaction, scaffolding, and feedback rather than answer delivery alone. More recently, this perspective has been extended to LLM-based tutoring through datasets and evaluation frameworks that explicitly model pedagogical dialogue [Macina et al., 2023, Sharma et al., 2026].

A growing line of work focuses on **tutoring-oriented dialogue datasets**. MathDial introduced a large tutoring dataset grounded in math reasoning problems and annotated with rich pedagogical properties, showing that such data can be used to fine-tune models to act more like tutors than solvers [Macina et al., 2023]. ConvoLearn similarly frames tutoring through explicit pedagogical dimensions such as cognitive engagement, formative assessment, and metacognition, though it is semi-synthetic rather than based on naturally occurring student use



(a) TF-IDF + LogReg (b) Sentence-MLP (c) TextCNN

Figure 1: Confusion matrices for goal prediction (5-class, test $n = 17$).



(a) TF-IDF + LogReg (b) Sentence-MLP (c) TextCNN

Figure 2: Confusion matrices for satisfaction prediction (3-class, LOOCV $n = 64$).

[Sharma et al., 2026]. These datasets are relevant because they show how tutoring behavior can be operationalized and modeled, but they differ from our project in that we focus on authentic student–LLM interactions.

A second line of work studies **real student–AI interactions**. StudyChat analyzes real student dialogues with ChatGPT in an AI course, providing evidence about how students actually ask questions and follow up on model responses [McNichols et al., 2025]. Related work has also examined AI tutor use in real courses at scale: Russell et al. study interaction patterns with an AI tutor in a large introductory STEM course and relate these patterns to course performance [Russell et al., 2025]. Beyond STEM tutoring, Han et al. analyze student–ChatGPT dialogue in EFL writing education, showing that authentic student–AI interactions can also reveal intent patterns and satisfaction signals in writing-support contexts [Han et al., 2023]. Together, these works broaden the landscape from controlled tutoring setups to real educational use.

A third line of work asks how dialogue data can be used to **improve tutors or model learning**. Scarlatos et al. train LLM-based tutors to maximize downstream student correctness while maintaining pedagogical quality, showing that dialogue-level optimization can improve tutor responses [Scarlatos et al., 2025b]. In a related direction, Scarlatos et al. also show that tutor–student dialogues can support knowledge tracing, using LLMs to identify skills and student correctness across turns and then modeling student understanding over time [Scarlatos et al., 2025a]. These papers are important for our project because they suggest that tutoring dialogues are useful not only for descriptive analysis, but also for downstream tutor improvement and student modeling.

Finally, recent work has begun to examine **engagement and risks**. Bai et al. study GPTutor, a generative-AI-powered tutoring system, and analyze behavioral, cognitive, and emotional engagement in higher education [Bai et al., 2025]. At the same time, Bastani et al. show that generative AI without sufficient guardrails can harm learning, as students may use the system as a crutch and perform worse once support is removed [Bastani et al., 2025]. These findings motivate our interest not only in whether LLM tutors are helpful, but in what kinds of feedback and interaction patterns are pedagogically productive.

Building on these strands of work, our project focuses specifically on authentic, open-ended student–LLM tutoring dialogues, with fine-grained annotation of tutor feedback and student follow-up behavior. Compared with prior work, our goal is to connect real usage data with pedagogical analysis and scalable annotation through few-shot prompting.

5 Conclusion

In this project, we build a dataset of authentic student–LLM tutoring dialogues focused primarily on Computer Science and Mathematics learning contexts. Motivated by the growing use of LLMs as informal educational tools, our project aims to better understand how students naturally interact with these systems and whether LLMs provide meaningful pedagogical support beyond simply generating correct answers.

Our main research questions focus on how students use LLMs for learning, what tutoring strategies commonly appear in model responses, how prompt quality relates to response quality, and how LLMs respond to student confusion across multiple turns. To study these questions, we developed an annotation and evaluation framework for analyzing learning scenarios, learning goals, pedagogical guidance, and tutoring quality in student–LLM interactions.

Although our analyses remain preliminary, the project demonstrates the feasibility of studying authentic tutoring interactions at scale and highlights the importance of evaluating pedagogical quality in addition to correctness. Our dataset also provides a structured foundation for future computational analysis of tutoring behavior, student engagement, and multi-turn educational interactions.

One limitation of the current project is the relatively small dataset size, which limits the reliability of large-scale supervised learning experiments and broader statistical conclusions. In addition, the dataset primarily contains English-language dialogues collected from students at Columbia University and Barnard College. As a result, the current dataset may not fully capture tutoring behaviors and learning patterns across different educational, linguistic, and cultural contexts. Annotation subjectivity also remains an important challenge, particularly when evaluating pedagogical usefulness and guidance quality.

Future work could expand the dataset to include more diverse student populations, languages, and educational environments, while also increasing the number of human annotators and measuring inter-annotator agreement. Additional directions include comparing different LLM systems more systematically, conducting larger-scale quantitative analyses, and exploring prompting or fine-tuning methods for improving pedagogical tutoring behavior.

Overall, this project contributes a real-world dataset and pedagogically grounded analysis framework for studying LLM tutoring behavior in authentic educational settings, laying a foundation for future research on building more effective, reliable, and student-centered AI tutoring systems.

6 Contribution Statement

Jessica Liu managed the report writing, conducted preliminary annotation experiments, performed data analytics, and helped with data collection.

Lexus Chen managed dataset collection, organized and cleaned the dialogue data, and contributed to report writing and data analytics.

Stella Wang implemented the prompting setup, designed the annotation scheme and evaluation guidelines, contributed to methodology framing and polishing the report.

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